



**The University of Azad Jammu and Kashmir,
Muzaffarabad**

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Roll No: 2024-SE-06

Course Title: Database System

Course Code: SE-2204

Lab No : 08

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Department Of Software Engineering

✓ Joins

1. INTRODUCTION

This lab was performed to understand SQL JOIN operations using MySQL in XAMPP. Different types of joins including INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, and multi-table joins were used to retrieve related data from multiple tables in the company schema.

2. OBJECTIVE

The objective of this lab was to practice SQL JOIN operations and retrieve related records from multiple database tables.

3. Create Database



4. Company schema tables created successfully.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ assignment	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KIB	-
☐ department	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 KIB	-
☐ employee	Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_general_ci	32.0 KIB	-
☐ project	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	32.0 KIB	-
4 tables	Sum	20	InnoDB	utf8mb4_general_ci	112.0 KIB	0 B

☐ Check all With selected:

Sample data inserted into all company schema tables.

Joins Part A

`SELECT * FROM `assignment``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: N

Extra options

				EmpID	ProjectID	WeeklyHours			
☐		Edit		Copy		Delete	1	1	20
☐		Edit		Copy		Delete	2	3	15
☐		Edit		Copy		Delete	3	2	18
☐		Edit		Copy		Delete	6	1	25
☐		Edit		Copy		Delete	7	4	12

☐ Check all | With selected: Edit Copy Delete Export

`SELECT * FROM `department``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key:

Extra options

				DeptID	DeptName	Location			
☐		Edit		Copy		Delete	1	Engineering	Lahore
☐		Edit		Copy		Delete	2	HR	Karachi
☐		Edit		Copy		Delete	3	Marketing	Islamabad
☐		Edit		Copy		Delete	4	Finance	Lahore

☐ Check all | With selected: Edit Copy Delete Export

Joins Part A

`SELECT * FROM `employee``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: | Filter rows: | Sort by key:

Extra options

				EmplID	EmpName	Salary	City	DeptID	ManagerID
☐				1	Ali Khan	95000.00	Lahore	1	NULL
☐				2	Sara Ahmed	72000.00	Karachi	2	1
☐				3	Bilal Hussain	68000.00	Islamabad	3	1
☐				4	Maham Noor	88000.00	Lahore	2	1
☐				5	Usman Tariq	120000.00	Karachi	1	NULL
☐				6	Areeba Khan	85000.00	Lahore	1	5
☐				7	Daniyal Raza	76000.00	Islamabad	4	5

☐ Check all | With selected: Edit Copy Delete Export

`SELECT * FROM `project``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: | Filter rows: | Sort by key:

Extra options

				ProjectID	ProjectName	DeptID	StartDate
☐				1	Mobile App	1	2024-01-10
☐				2	Website Redesign	3	2024-03-15
☐				3	HR System	2	2023-11-20
☐				4	Finance Tracker	NULL	2024-02-05

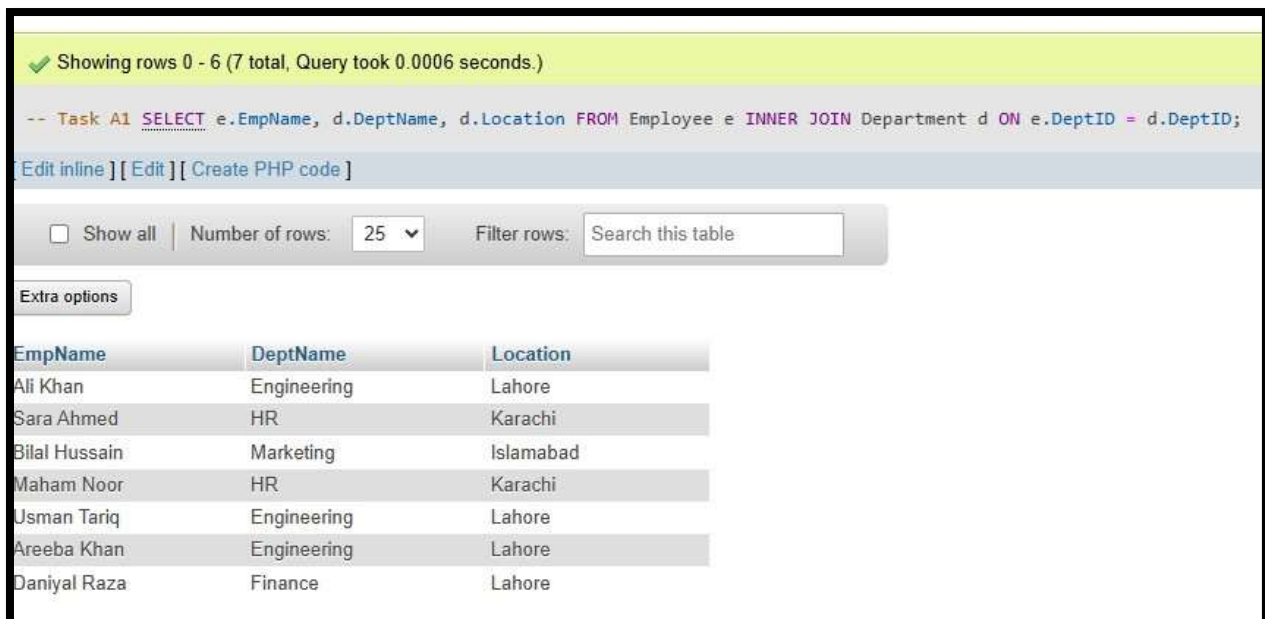
☐ Check all | With selected: Edit Copy Delete Export

5. Part A Queries

➤ TaskA1

- This query lists every employee with their department name and department location using INNER JOIN.

```
SELECT e.EmpName, d.DeptName, d.Location
FROM Employee
INNER JOIN Department d
ON e.DeptID = d.DeptID;
```



The screenshot shows a database query interface. At the top, a green status bar indicates 'Showing rows 0 - 6 (7 total, Query took 0.0006 seconds.)'. Below this, the SQL query is displayed: `-- Task A1 SELECT e.EmpName, d.DeptName, d.Location FROM Employee e INNER JOIN Department d ON e.DeptID = d.DeptID;`. There are links for 'Edit inline', 'Edit', and 'Create PHP code'. Below the query, there are controls for 'Show all', 'Number of rows' (set to 25), and a 'Filter rows' search box. An 'Extra options' button is also present. The main part of the screenshot is a table with three columns: 'EmpName', 'DeptName', and 'Location'. The table contains seven rows of data.

EmpName	DeptName	Location
Ali Khan	Engineering	Lahore
Sara Ahmed	HR	Karachi
Bilal Hussain	Marketing	Islamabad
Maham Noor	HR	Karachi
Usman Tariq	Engineering	Lahore
Areeba Khan	Engineering	Lahore
Daniyal Raza	Finance	Lahore

Task A1 showing employees with their department names and locations.

➤ TaskA2

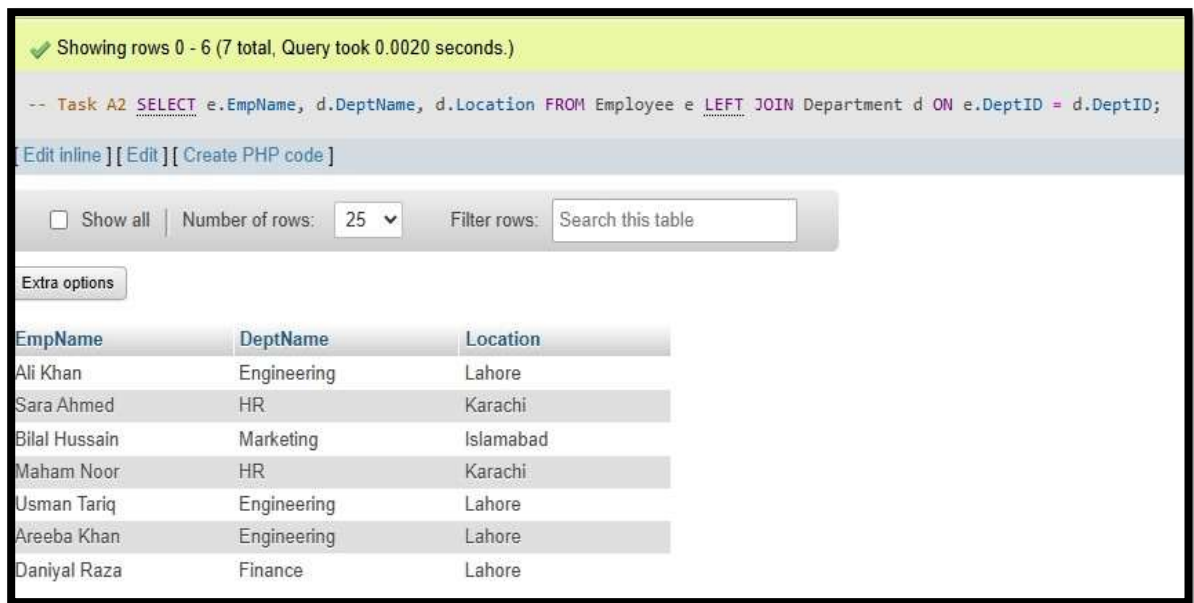
- This query displays all employees including those whose department ID is NULL using LEFT JOIN.

```
SELECT e.EmpName, d.DeptName, d.Location
FROM Employee e
```

Joins Part A

LEFT JOIN Department d

ON e.DeptID = d.DeptID



Showing rows 0 - 6 (7 total, Query took 0.0020 seconds.)

-- Task A2 `SELECT e.EmpName, d.DeptName, d.Location FROM Employee e LEFT JOIN Department d ON e.DeptID = d.DeptID;`

[Edit inline](#) [\[Edit\]](#) [\[Create PHP code\]](#)

☐ Show all | Number of rows: 25 ▼ | Filter rows:

Extra options

EmpName	DeptName	Location
Ali Khan	Engineering	Lahore
Sara Ahmed	HR	Karachi
Bilal Hussain	Marketing	Islamabad
Maham Noor	HR	Karachi
Usman Tariq	Engineering	Lahore
Areeba Khan	Engineering	Lahore
Daniyal Raza	Finance	Lahore

Task A2 showing all employees with department details.

➤ TaskA3

- This query lists every department along with employee names. Departments without employees also appear in the result.

SELECT d.DeptName, e.EmpName
FROM Department d
LEFT JOIN Employee e
ON d.DeptID = e.DeptID;

Joins Part A

✓ Showing rows 0 - 6 (7 total, Query took 0.0008 seconds.)

```
-- Task A3 SELECT d.DeptName, e.EmpName FROM Department d LEFT JOIN Employee e ON d.DeptID = e.DeptID;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

DeptName	EmpName
Engineering	Ali Khan
Engineering	Usman Tariq
Engineering	Areeba Khan
HR	Sara Ahmed
HR	Maham Noor
Marketing	Bilal Hussain
Finance	Daniyal Raza

Task A3 showing departments and their employees.

➤ Task A4

- This query displays every project with department name and location including projects without departments.

```
SELECT p.ProjectName, d.DeptName, d.Location  
FROM Project p LEFT JOIN Department d  
ON p.DeptID = d.DeptID;
```

✓ Showing rows 0 - 3 (4 total, Query took 0.0007 seconds.)

```
-- Task A4 SELECT p.ProjectName, d.DeptName, d.Location FROM Project p LEFT JOIN Department d ON p.DeptID = d.DeptID;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

ProjectName	DeptName	Location
Mobile App	Engineering	Lahore
Website Redesign	Marketing	Islamabad
HR System	HR	Karachi
Finance Tracker	NULL	NULL

Task A4 showing projects with department information.

➤ **TaskA5**

- This query finds employees who are not assigned to any project using LEFT JOIN and IS NULL.

```
SELECT e.EmpName
FROM Employee e
LEFT JOIN Assignment a
ON e.EmpID = a.EmpID
WHERE a.ProjectID IS NULL;
```



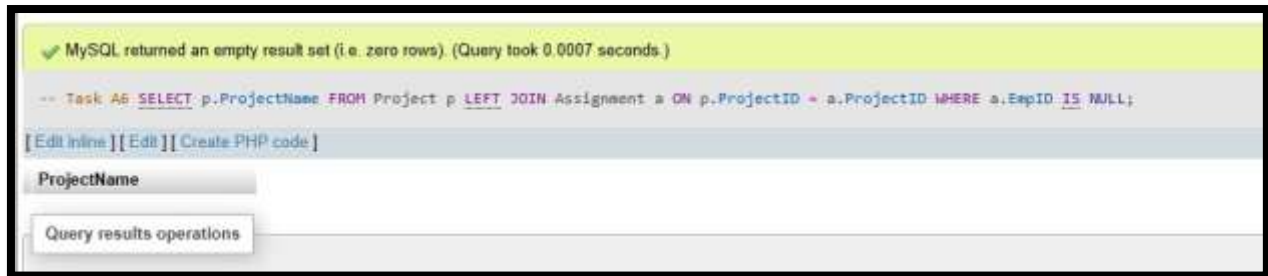
Task A5 showing employees without project assignments.

➤ **TaskA6**

- This query lists projects that currently have no employee assignments.

```
SELECT p.ProjectName
FROM Project p
LEFT JOIN Assignment a
ON p.ProjectID = a.ProjectID
WHERE a.EmpID IS NULL;
```

Joins Part A



Task A6 showing projects without assignments.

➤ TaskA7

- This query displays employees in the Engineering department sorted by salary in descending order.

SELECT e.EmpName, e.Salary

FROM Employee e

INNER JOIN Department d

ON e.DeptID = d.DeptID

WHERE d.DeptName = 'Engineering' ORDER BY e.Salary DESC;



Task A7 showing Engineering employees sorted by salary.

➤ **TaskA8**

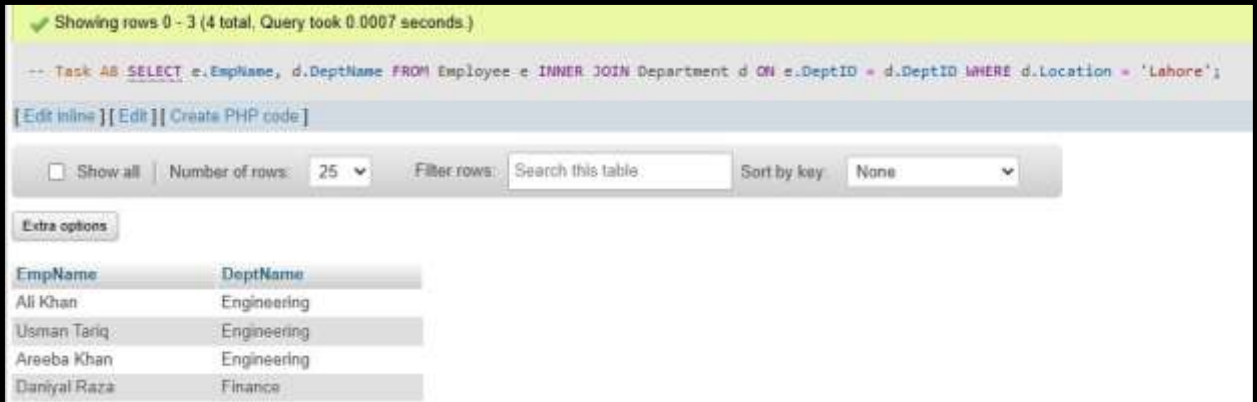
- This query lists employees working in departments located in Lahore.

```
SELECT e.EmpName, d.DeptName
```

```
FROM Employee e
```

```
INNER JOIN Department d ON e.DeptID = d.DeptID
```

```
WHERE d.Location = 'Lahore';
```



The screenshot shows a database query interface. At the top, a status bar indicates 'Showing rows 0 - 3 (4 total, Query took 0.0007 seconds)'. Below this, the SQL query is displayed: `-- Task A8 SELECT e.EmpName, d.DeptName FROM Employee e INNER JOIN Department d ON e.DeptID = d.DeptID WHERE d.Location = 'Lahore';`. There are links for '[Edit inline]', '[Edit]', and 'Create PHP code'. Below the query, there are controls for 'Show all', 'Number of rows' (set to 25), 'Filter rows' (with a search box), and 'Sort by key' (set to None). An 'Extra options' button is also present. The result is displayed as a table with two columns: 'EmpName' and 'DeptName'. The data rows are: Ali Khan (Engineering), Usman Tariq (Engineering), Areeba Khan (Engineering), and Daniyal Raza (Finance).

EmpName	DeptName
Ali Khan	Engineering
Usman Tariq	Engineering
Areeba Khan	Engineering
Daniyal Raza	Finance

Task A8 showing employees in Lahore-based departments.

➤ **TaskA9**

- This query displays every department with the total number of employees working in each department.

```
SELECT d.DeptName, COUNT(e.EmpID) AS EmployeeCount
```

```
FROM Department d
```

```
LEFT JOIN Employee e
```

```
ON d.DeptID = e.DeptID GROUP BY d.DeptName;
```

Joins Part A



The screenshot shows a database query interface. At the top, a status bar indicates 'Showing rows 0 - 3 (4 total. Query took 0.0006 seconds.)'. Below this, the SQL query is displayed: `-- Task A9: SELECT d.DeptName, COUNT(e.EmpID) AS EmployeeCount FROM Department d LEFT JOIN Employee e ON d.DeptID = e.DeptID GROUP BY d.DeptName;`. Below the query, there are controls for 'Show all', 'Number of rows' (set to 25), and a 'Filter rows' search box. An 'Extra options' button is also present. The query result is displayed in a table with two columns: 'DeptName' and 'EmployeeCount'. The data rows are: Engineering (3), Finance (1), HR (2), and Marketing (1).

DeptName	EmployeeCount
Engineering	3
Finance	1
HR	2
Marketing	1

Task A9 showing employee count in each department.

➤ TaskA10

- This query produces a FULL OUTER JOIN result of Employee and Department tables using UNION.

```
SELECT e.EmpName, d.DeptName
FROM Employee e
LEFT JOIN Department d
ON e.DeptID = d.DeptID
UNION SELECT e.EmpName, d.DeptName
FROM Employee e
RIGHT JOIN Department d
ON e.DeptID = d.DeptID;
```

Joins Part A



Showing rows 0 - 6 (7 total, Query took 0.0015 seconds)

```
-- Task A10 SELECT e.EmpName, d.DeptName FROM Employee e LEFT JOIN Department d ON e.DeptID = d.DeptID UNION SELECT e.EmpName, d.DeptName FROM Employee e RIGHT JOIN Department d ON e.DeptID = d.DeptID;
```

Extra options

EmpName	DeptName
Ali Khan	Engineering
Sara Ahmed	HR
Bilal Hussain	Marketing
Maham Noor	HR
Usman Tariq	Engineering
Areeba Khan	Engineering
Daniyal Raza	Finance

Task A10 showing FULL OUTER JOIN using UNION.

6. Conclusion

In this lab, different SQL JOIN operations were performed successfully using MySQL in XAMPP. INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, and multi-table joins helped in retrieving related information from multiple tables. The lab improved understanding of relational database concepts and SQL query writing.